

数列专题习题册 (中英双语版) 答案与解析

一、选择题 (I. Multiple Choice Questions)

题号	答案	解析 (Explanation)	英文解析 (English Explanation)
1	A	等差中项公式:若 a, A, b 成等差数列, 则 $A = \frac{a+b}{2}$ 。题目中两数为 $\sqrt{5}$ 和 $\sqrt{5}$, 代入公式得: $A = \sqrt{5}$, 选 A。	Arithmetic mean formula: If a, A, b are in arithmetic sequence, then $A = \frac{a+b}{2}$. The two numbers in the question are $\sqrt{5}$ and $\sqrt{5}$. Substituting into the formula: $A = \sqrt{5}$, so choose A.
2	B	等差数列通项公式: $a_n = a_1 + (n-1)d$ 。	Arithmetic sequence general term formula: $a_n = a_1 + (n-1)d$.
3	C	等比中项公式:若 a, G, b 成等比数列, 则 $G = \pm \sqrt{ab}$ 。代入 $a = 4$, $b = 16$, 得 $G = \pm \sqrt{4 \times 16} = \pm 8$, 选 C。	Geometric mean formula: If a, G, b are in geometric sequence, then $G = \pm \sqrt{ab}$. Substitute $a = 4$ and $b = 16$, we get $G = \pm \sqrt{4 \times 16} = \pm 8$, so choose C.
4	D	等比数列公比公式: $q = \frac{a_{n+1}}{a_n}$ 。已知 $a_2 = 6$, $a_3 = 12$, 则 $q = \frac{12}{6} = 2$, 选 D。	Geometric sequence common ratio formula: $q = \frac{a_{n+1}}{a_n}$. Given $a_2 = 6$ and $a_3 = 12$, then $q = \frac{12}{6} = 2$,

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			so choose D.
5	A	等差数列通项公式: $a_n = a_1 + (n - 1)d$ 。已知 $a_1 = 3, d = 2, n = 5$, 则 $a_5 = 3 + 4 \times 2 = 11$, 选 A。	Arithmetic sequence general term formula: $a_n = a_1 + (n - 1)d$. Given $a_1 = 3, d = 2, n = 5$, then $a_5 = 3 + 4 \times 2 = 11$, so choose A.
6	A	等差数列前 n 项和公式: $S_n = \frac{n(a_1 + a_n)}{2}$ 。已知 $a_3 = 7, S_3 = 15$, 则 $15 = \frac{3(a_1 + 7)}{2}$, 解得 $a_1 = 3$ 。由通项公式 $a_3 = a_1 + 2d$, 得 $7 = 3 + 2d$, $d = 2$, 选 A。	Arithmetic sequence sum of first n terms formula: $S_n = \frac{n(a_1 + a_n)}{2}$. Given $a_3 = 7$ and $S_3 = 15$, then $15 = \frac{3(a_1 + 7)}{2}$, so $a_1 = 3$. From the general term formula $a_3 = a_1 + 2d$, we get $7 = 3 + 2d, d = 2$, so choose A.
7	B	等差数列通项公式推论: $a_m = a_n + (m - n)d$ 。已知 $a_3 = 6, a_7 = 22$, 则 $22 = 6 + 4d$, 解得 $d = 4$, 选 B。	Arithmetic sequence general term formula corollary: $a_m = a_n + (m - n)d$. Given $a_3 = 6$ and $a_7 = 22$, then $22 = 6 + 4d$, so $d = 4$, choose B.
8	A	等比数列前 n 项和公式: $S_n = a_1(1 + q + q^2 + \dots + q^{n-1})$ 。已知 $a_1 = 3, S_3 = 39$, 则	Geometric sequence sum of first n terms formula: $S_n = a_1(1 + q + q^2 + \dots + q^{n-1})$. Given $a_1 = 3$ and

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		$3(1 + q + q^2) = 39$, 即 $q^2 + q - 12 = 0$ 。解得 $q = 3$ 或 $q = -4$, 因数列递增且 $a_1 > 0$, 故 $q = 3$, 选 A。	$S_3 = 39$, then $3(1 + q + q^2) = 39$, i.e., $q^2 + q - 12 = 0$. Solve for q : $q = 3$ or $q = -4$. Since the sequence is increasing and $a_1 > 0$, $q = 3$, so choose A.
9	C、D	等差数列定义: 后项 - 前项 = 常数(公差 d)。 - A: $(1 + \sqrt{2}) - 1 = \sqrt{2}$, $(1 + \sqrt{3}) - (1 + \sqrt{2}) = \sqrt{3} - \sqrt{2}$, 不相等, 非等差数列; - B: $4 - 1 = 3$, $16 - 4 = 12$, 不相等, 非等差数列; - C: $\lg 9 - \lg 3 = \lg 3$, $\lg 27 - \lg 9 = \lg 3$, 相等, 是等差数列; - D: $1 - (-4) = 5$, $6 - 1 = 5$, 相等, 是等差数列。 题目若为多选题, 选 C、D; 若为单选题, 推测题干疏漏, 优先选 C。	Arithmetic sequence definition: Subsequent term - Previous term = Constant (common difference d). - A: $(1 + \sqrt{2}) - 1 = \sqrt{2}$, $(1 + \sqrt{3}) - (1 + \sqrt{2}) = \sqrt{3} - \sqrt{2}$, not equal, not an arithmetic sequence; - B: $4 - 1 = 3$, $16 - 4 = 12$, not equal, not an arithmetic sequence; - C: $\lg 9 - \lg 3 = \lg 3$, $\lg 27 - \lg 9 = \lg 3$, equal, it is an arithmetic sequence; - D: $1 - (-4) = 5$, $6 - 1 = 5$, equal, it is an arithmetic sequence. If the question is a multiple-choice question, choose C and D; if it is a single-choice

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			question, it is speculated that there is an oversight in the question stem, and C is preferred.
10	B	等比数列通项公式: $a_n = a_1 q^{n-1}$ 。已知 $a_1 = 4$, $q = \frac{1}{3}$, $n = 3$, 则 $a_3 = 4 \times \left(\frac{1}{3}\right)^2 = \frac{4}{9}$, 选 B。	Geometric sequence general term formula: $a_n = a_1 q^{n-1}$. Given $a_1 = 4$, $q = \frac{1}{3}$, $n = 3$, then $a_3 = 4 \times \left(\frac{1}{3}\right)^2 = \frac{4}{9}$, so choose B.

二、填空题 (II. Fill-in-the-Blank Questions)

题号	答案	解析 (Explanation)	英文解析 (English Explanation)
1	$\frac{4}{7}$	由通项公式 $a_n = \frac{a_{n-1}}{2^{n-1}}$ ($n \in \mathbb{N}^*$), $a_2 = 4$, 则 $a_3 = \frac{a_2}{2^{3-1}} = \frac{4}{8-1} = \frac{4}{7}$ 。	From the general term formula $a_n = \frac{a_{n-1}}{2^{n-1}}$ ($n \in \mathbb{N}^*$) and $a_2 = 4$, then $a_3 = \frac{a_2}{2^{3-1}} = \frac{4}{8-1} = \frac{4}{7}$.
2	$\frac{9}{2}$	化简通项公式: $a_n = (-3)^2 \cdot \frac{1}{n} = 9 \cdot \frac{1}{n}$, 则 $a_2 = \frac{9}{2}$ 。	Simplify the general term formula: $a_n = (-3)^2 \cdot \frac{1}{n} = 9 \cdot \frac{1}{n}$, then $a_2 = \frac{9}{2}$.
3	3; 11	由递推公式 $a_n = 3a_{n-1} + 2$ ($n > 1$): $a_2 = 3 \times \frac{1}{3} + 2 = 1 + 2 = 3$; $a_3 = 3 \times 3 + 2 = 9 + 2 =$	From the recursive formula $a_n = 3a_{n-1} + 2$ ($n > 1$): $a_2 = 3 \times \frac{1}{3} + 2 = 1 + 2 = 3$; $a_3 = 3 \times 3 + 2 =$

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		11。	$9 + 2 = 11.$
4	3	数列项与前 n 项和关系: $a_n = S_n - S_{n-1} (n \geq 2)$ 。 $a_3 = S_3 - S_2 = (3 \times 3 - 5) - (3 \times 2 - 5) = 4 - 1 = 3$ 。	Relationship between sequence terms and sum of first n terms: $a_n = S_n - S_{n-1} (n \geq 2)$. $a_3 = S_3 - S_2 = (3 \times 3 - 5) - (3 \times 2 - 5) = 4 - 1 = 3.$
5	7; 14; 33; 54	由通项公式 $a_n = n^3 + 6$: $a_1 = 1^3 + 6 = 7$; $a_2 = 2^3 + 6 = 14$; $a_3 = 3^3 + 6 = 33$; $S_3 = 7 + 14 + 33 = 54$ 。	From the general term formula $a_n = n^3 + 6$: $a_1 = 1^3 + 6 = 7$; $a_2 = 2^3 + 6 = 14$; $a_3 = 3^3 + 6 = 33$; $S_3 = 7 + 14 + 33 = 54.$
6	12	等差中项公式: $A = \frac{11+13}{2} = 12$ 。	Arithmetic mean formula: $A = \frac{11+13}{2} = 12.$
7	-20	等差数列通项公式: $a_7 = a_1 + 6d = 4 + 6 \times (-4) = 4 - 24 = -20$ 。	Arithmetic sequence general term formula: $a_7 = a_1 + 6d = 4 + 6 \times (-4) = 4 - 24 = -20.$
8	3	等差数列通项公式推论: $a_7 = a_1 + 6d$, 则 $20 = 2 + 6d$, 解得 $d = 3$ 。	Arithmetic sequence general term formula corollary: $a_7 = a_1 + 6d$, then $20 = 2 + 6d$, so $d = 3.$
9	$\frac{7}{2}$; 4	通项公式推论: $a_5 = a_1 + 4d$, 则 $11 = -3 +$	General term formula corollary: $a_5 = a_1 + 4d$, then

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		$4d$, 解得 $d = \frac{7}{2}$; $a_3 = a_1 + 2d = -3 + 2 \times \frac{7}{2} = 4$ 。	$11 = -3 + 4d$, so $d = \frac{7}{2}$; $a_3 = a_1 + 2d = -3 + 2 \times \frac{7}{2} = 4$.
10	1	通项公式: $a_7 = a_1 + 6d$, 则 $-11 = a_1 + 6 \times (-2)$, 解得 $a_1 = 1$ 。	General term formula: $a_7 = a_1 + 6d$, then $-11 = a_1 + 6 \times (-2)$, so $a_1 = 1$.
11	28	等比数列前 n 项和公式: $S_n = \frac{a_1(1-q^n)}{1-q}$. $S_3 = \frac{4(1-(-3)^3)}{1-(-3)} = \frac{4 \times 28}{4} = 28$ 。	Geometric sequence sum of first n terms formula: $S_n = \frac{a_1(1-q^n)}{1-q}$. $S_3 = \frac{4(1-(-3)^3)}{1-(-3)} = \frac{4 \times 28}{4} = 28$.
12	2	公比公式: $q = \frac{a_4}{a_3} = \frac{10}{5} = 2$ 。	Common ratio formula: $q = \frac{a_4}{a_3} = \frac{10}{5} = 2$.
13	$\hat{A} \pm 2$; $\hat{A} \pm 5$	通项公式推论: $a_4 = a_2 q^2$, 则 $40 = 10q^2$, 解得 $q = \hat{A} \pm 2$; $a_1 = \frac{a_2}{q} = \frac{10}{\hat{A} \pm 2} = \hat{A} \pm 5$ 。	General term formula corollary: $a_4 = a_2 q^2$, then $40 = 10q^2$, so $q = \hat{A} \pm 2$; $a_1 = \frac{a_2}{q} = \frac{10}{\hat{A} \pm 2} = \hat{A} \pm 5$.
14	1	通项公式: $a_3 = a_1 q^2$, 则 $\frac{1}{9} = a_1 \times \left(\frac{1}{3}\right)^2$, 解得 $a_1 = 1$ 。	General term formula: $a_3 = a_1 q^2$, then $\frac{1}{9} = a_1 \times \left(\frac{1}{3}\right)^2$, so $a_1 = 1$.
15	$\hat{A} \pm \frac{\sqrt{6}}{3}$	等比中项公式: $G = \hat{A} \pm \sqrt{\frac{\sqrt{6}}{3} \times \frac{\sqrt{6}}{3}} = \hat{A} \pm \sqrt{\frac{6}{9}} = \hat{A} \pm \frac{\sqrt{6}}{3}$ 。	Geometric mean formula: $G = \hat{A} \pm \sqrt{\frac{\sqrt{6}}{3} \times \frac{\sqrt{6}}{3}} = \hat{A} \pm \sqrt{\frac{6}{9}} = \hat{A} \pm \frac{\sqrt{6}}{3}$.

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16	$\hat{A} \pm 6$	等比中项公式: $G = \hat{A} \pm \sqrt{3 \times 12} = \hat{A} \pm \sqrt{36} = \hat{A} \pm 6$ 。	Geometric mean formula: $G = \hat{A} \pm \sqrt{3 \times 12} = \hat{A} \pm \sqrt{36} = \hat{A} \pm 6$.
17	-10; 4	通项公式: $a_8 = a_1 + 7d$, 则 $11 = a_1 + 7 \times 3$, 解得 $a_1 = -10$; 前 n 项和: $S_8 = \frac{8(a_1 + a_8)}{2} = 4 \times (-10 + 11) = 4$ 。	General term formula: $a_8 = a_1 + 7d$, then $11 = a_1 + 7 \times 3$, so $a_1 = -10$; Sum of first n terms: $S_8 = \frac{8(a_1 + a_8)}{2} = 4 \times (-10 + 11) = 4$.
18	-4	通项公式: $a_7 = a_1 + 6d = -6 + 6 \times \frac{1}{3} = -6 + 2 = -4$ 。	General term formula: $a_7 = a_1 + 6d = -6 + 6 \times \frac{1}{3} = -6 + 2 = -4$.
19	1	通项公式推论: $a_8 = a_3 + 5d$, 则 $15 = 5 + 5d$, 解得 $d = 2$; $a_1 = a_3 - 2d = 5 - 4 = 1$ 。	General term formula corollary: $a_8 = a_3 + 5d$, then $15 = 5 + 5d$, so $d = 2$; $a_1 = a_3 - 2d = 5 - 4 = 1$.
20	20	先求公差: $d = \frac{a_4 - a_1}{3} = \frac{8 - 2}{3} = 2$; 前 n 项和: $S_4 = \frac{4(a_1 + a_4)}{2} = 2 \times (2 + 8) = 20$ 。	First find the common difference: $d = \frac{a_4 - a_1}{3} = \frac{8 - 2}{3} = 2$; Sum of first n terms: $S_4 = \frac{4(a_1 + a_4)}{2} = 2 \times (2 + 8) = 20$.
21	19	通项公式推论: $a_6 = a_2 + 4d$, 则 $11 = 3 + 4d$, 解得 $d = 2$; $a_{10} = a_6 + 4d = 11 + 8 =$	General term formula corollary: $a_6 = a_2 + 4d$, then $11 = 3 + 4d$, so $d = 2$; $a_{10} = a_6 + 4d = 11 + 8 =$

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		19。	19.
22	3	公比公式: $q = \frac{a_4}{a_3} = \frac{15}{5} = 3$ 。	Common ratio formula: $q = \frac{a_4}{a_3} = \frac{15}{5} = 3$.
23	3	通项公式: $a_4 = a_1 q^3$, 则 $9 = \frac{1}{3} q^3$, 解得 $q^3 = 27$, $q = 3$ 。	General term formula: $a_4 = a_1 q^3$, then $9 = \frac{1}{3} q^3$, so $q^3 = 27$, $q = 3$.
24	2	公比公式: $q = \frac{a_6}{a_5} = \frac{6}{3} = 2$ 。	Common ratio formula: $q = \frac{a_6}{a_5} = \frac{6}{3} = 2$.
25	62	前 3 项和: $S_3 = a_1 + a_1 q + a_1 q^2 = 2 + 10 + 50 = 62$ 。	Sum of first 3 terms: $S_3 = a_1 + a_1 q + a_1 q^2 = 2 + 10 + 50 = 62$.
26	$\hat{A} \pm 2$; 2	通项公式推论: $a_5 = a_3 q^2$, 则 $32 = 8q^2$, 解得 $q = \hat{A} \pm 2$; $a_1 = \frac{a_3}{q^2} = \frac{8}{4} = 2$ 。	General term formula corollary: $a_5 = a_3 q^2$, then $32 = 8q^2$, so $q = \hat{A} \pm 2$; $a_1 = \frac{a_3}{q^2} = \frac{8}{4} = 2$.
27	8	等差中项公式: $A = \frac{7+9}{2} = 8$ 。	Arithmetic mean formula: $A = \frac{7+9}{2} = 8$.
28	$\frac{5\sqrt{7}}{2}$	等差中项公式: $A = \frac{\sqrt{7}+4\sqrt{7}}{2} = \frac{5\sqrt{7}}{2}$ 。	Arithmetic mean formula: $A = \frac{\sqrt{7}+4\sqrt{7}}{2} = \frac{5\sqrt{7}}{2}$.
29	$2a$	等差中项公式: $A = \frac{(2a+b)+(2a-b)}{2} = \frac{4a}{2} = 2a$ 。	Arithmetic mean formula: $A = \frac{(2a+b)+(2a-b)}{2} = \frac{4a}{2} = 2a$.

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30	$\hat{A} \pm 5\sqrt{3}$	等比中项公式: $G = \hat{A} \pm \sqrt{5 \times 15} = \hat{A} \pm \sqrt{75} = \hat{A} \pm 5\sqrt{3}$ 。	Geometric mean formula: $G = \hat{A} \pm \sqrt{5 \times 15} = \hat{A} \pm \sqrt{75} = \hat{A} \pm 5\sqrt{3}$.
31	$\hat{A} \pm \sqrt{2}$	等比中项公式: $G = \hat{A} \pm \sqrt{\frac{\sqrt{8}}{2} \times \frac{\sqrt{8}}{2}} = \hat{A} \pm \sqrt{\frac{8}{4}} = \hat{A} \pm \sqrt{2}$ 。	Geometric mean formula: $G = \hat{A} \pm \sqrt{\frac{\sqrt{8}}{2} \times \frac{\sqrt{8}}{2}} = \hat{A} \pm \sqrt{\frac{8}{4}} = \hat{A} \pm \sqrt{2}$.
32	$9; \hat{A} \pm 3\sqrt{5}$	等差中项: $A = \frac{3+15}{2} = 9$; 等比中项: $G = \hat{A} \pm \sqrt{3 \times 15} = \hat{A} \pm \sqrt{45} = \hat{A} \pm 3\sqrt{5}$ 。	Arithmetic mean: $A = \frac{3+15}{2} = 9$; Geometric mean: $G = \hat{A} \pm \sqrt{3 \times 15} = \hat{A} \pm \sqrt{45} = \hat{A} \pm 3\sqrt{5}$.
33	7	通项公式: $a_3 = 3^2 - 2 = 9 - 2 = 7$ 。	General term formula: $a_3 = 3^2 - 2 = 9 - 2 = 7$.
34	6	项与前 n 项和关系: $a_2 = S_2 - S_1 = 6 \times (2 + 3) - 6 \times (1 + 3) = 30 - 24 = 6$ 。	Relationship between term and sum of first n terms: $a_2 = S_2 - S_1 = 6 \times (2 + 3) - 6 \times (1 + 3) = 30 - 24 = 6$.

三、计算题 (III. Calculation Questions)

1. 答案 (Answer)

首项 $a_1 = 3$, 前 4 项和 $S_4 = 30$ 。First term $a_1 = 3$, sum of the first 4 terms $S_4 = 30$.

解析 (Explanation)

1. 等差数列通项公式推论: $a_m = a_n + (m - n)d$ 。已知 $a_3 = 9$, $a_7 = 21$, 则: $a_7 = a_3 + 4d \rightarrow 21 = 9 + 4d \rightarrow$ 解得 $d = 3$ 。2. 求首项 a_1 : 由 $a_3 = a_1 + 2d$, 代入 $a_3 = 9$, $d = 3$: $9 = a_1 + 2 \times 3 \rightarrow$ 解得 $a_1 = 3$ 。3. 求前 4 项和 S_4 : 前 n 项和公式 $S_n = \frac{n(a_1 + a_n)}{2}$, 先求 $a_4 = a_1 + 3d = 3 + 9 = 12$: $S_4 = \frac{4 \times (3 + 12)}{2} = 2 \times 15 = 30$ 。

英文解析 (English Explanation)

1. Arithmetic sequence general term formula corollary: $a_m = a_n + (m - n)d$. Given $a_3 = 9$ and $a_7 = 21$, then: $a_7 = a_3 + 4d \rightarrow 21 = 9 + 4d \rightarrow$ Solve for $d = 3$. 2. Find the first term a_1 : From $a_3 = a_1 + 2d$, substitute $a_3 = 9$ and $d = 3$: $9 = a_1 + 2 \times 3 \rightarrow$ Solve for $a_1 = 3$. 3. Find the sum of the first 4 terms S_4 : Sum of first n terms formula $S_n = \frac{n(a_1 + a_n)}{2}$. First find $a_4 = a_1 + 3d = 3 + 9 = 12$: $S_4 = \frac{4 \times (3 + 12)}{2} = 2 \times 15 = 30$.

2. 答案 (Answer)

首项 $a_1 = 4$, 公差 $d = 2$ 。First term $a_1 = 4$, common difference $d = 2$.

解析 (Explanation)

1. 列方程:- 由 $a_2 = 6$ 得: $a_1 + d = 6$ (通项公式 $a_2 = a_1 + d$); - 由 $S_5 = 40$ 得: $\frac{5(a_1 + a_5)}{2} = 40$ (前 n 项和公式), 化简为 $a_1 + a_5 = 16$ 。2. 由通项公式 $a_5 = a_1 + 4d$, 代入 $a_1 + a_5 = 16$: $a_1 + (a_1 + 4d) = 16 \rightarrow 2a_1 + 4d = 16 \rightarrow$ 化简为 $a_1 + 2d = 8$ 。3. 联立方程组: $\begin{cases} a_1 + d = 6 \\ a_1 + 2d = 8 \end{cases} \rightarrow$ 两式相减得 $d = 2$, 代入 $a_1 + d = 6$ 得 $a_1 =$

4。

英文解析 (English Explanation)

1. Set up equations:- From $a_2 = 6$: $a_1 + d = 6$ (general term formula $a_2 = a_1 + d$); - From $S_5 = 40$: $\frac{5(a_1+a_5)}{2} = 40$ (sum of first n terms formula), simplified to $a_1 + a_5 = 16$. 2. From the general term formula $a_5 = a_1 + 4d$, substitute into $a_1 + a_5 = 16$: $a_1 + (a_1 + 4d) = 16 \rightarrow 2a_1 + 4d = 16 \rightarrow$ Simplified to $a_1 + 2d = 8$. 3. Solve the system of equations: $\begin{cases} a_1 + d = 6 \\ a_1 + 2d = 8 \end{cases} \rightarrow$ Subtract the first equation from the second: $d = 2$, substitute into $a_1 + d = 6$ to get $a_1 = 4$.

3.答案 (Answer)

- (1) $a_1 = -5$, $d = 6$; (2) $a_7 = 31$; (3) $S_4 = 16$ 。 (1) $a_1 = -5$, $d = 6$; (2) $a_7 = 31$; (3) $S_4 = 16$ 。

解析 (Explanation)

- (1) 求 a_1 和 d : 通项公式推论 $a_6 = a_3 + 3d$, 已知 $a_3 = 7$, $a_6 = 25$: $25 = 7 + 3d \rightarrow$ 解得 $d = 6$; 由 $a_3 = a_1 + 2d$: $7 = a_1 + 2 \times 6 \rightarrow$ 解得 $a_1 = -5$ 。(2) 求 a_7 : $a_7 = a_6 + d = 25 + 6 = 31$ 。(3) 求 S_4 : 前 n 项和公式 $S_4 = \frac{4(a_1+a_4)}{2} = 2(a_1 + a_4)$; $a_4 = a_1 + 3d = -5 + 18 = 13$; $S_4 = 2 \times (-5 + 13) = 2 \times 8 = 16$ 。

英文解析 (English Explanation)

- (1) Find a_1 and d : General term formula corollary $a_6 = a_3 + 3d$, given $a_3 = 7$ and $a_6 = 25$: $25 = 7 + 3d \rightarrow$

Solve for $d = 6$; From $a_3 = a_1 + 2d: 7 = a_1 + 2 \times 6 \rightarrow$ Solve for $a_1 = -5$ 。(2) Find $a_7: a_7 = a_6 + d = 25 + 6 = 31$ 。(3) Find S_4 : Sum of first n terms formula $S_4 = \frac{4(a_1 + a_4)}{2} = 2(a_1 + a_4)$; $a_4 = a_1 + 3d = -5 + 18 = 13$;
 $S_4 = 2 \times (-5 + 13) = 2 \times 8 = 16$ 。

4.答案 (Answer)

$$(1) a_1 = \frac{3}{8}, q = 2; (2) S_5 = \frac{93}{8}。 (1) a_1 = \frac{3}{8}, q = 2; (2) S_5 = \frac{93}{8}。$$

解析 (Explanation)

(1) 求 a_1 和 q : 等比数列通项公式推论 $a_7 = a_4 q^3$, 已知 $a_4 = 3, a_7 = 24: 24 = 3q^3 \rightarrow q^3 = 8 \rightarrow$ 解得 $q = 2$;

由 $a_4 = a_1 q^3: 3 = a_1 \times 2^3 \rightarrow$ 解得 $a_1 = \frac{3}{8}$ 。(2) 求 S_5 : 前 n 项和公式 $S_n = \frac{a_1(1-q^n)}{1-q}: S_5 = \frac{\frac{3}{8}(1-2^5)}{1-2} = \frac{\frac{3}{8} \times 31}{1} = \frac{93}{8}$ 。

英文解析 (English Explanation)

(1) Find a_1 and q : Geometric sequence general term formula corollary $a_7 = a_4 q^3$, given $a_4 = 3$ and $a_7 =$

$24: 24 = 3q^3 \rightarrow q^3 = 8 \rightarrow$ Solve for $q = 2$; From $a_4 = a_1 q^3: 3 = a_1 \times 2^3 \rightarrow$ Solve for $a_1 = \frac{3}{8}$ 。(2) Find

S_5 : Sum of first n terms formula $S_n = \frac{a_1(1-q^n)}{1-q}: S_5 = \frac{\frac{3}{8}(1-2^5)}{1-2} = \frac{\frac{3}{8} \times 31}{1} = \frac{93}{8}$ 。

5.答案 (Answer)

$$(1) A = 6; (2) d = -3; (3) S_4 = 18。 (1) A = 6; (2) d = -3; (3) S_4 = 18。$$

解析 (Explanation)

(1) 求等差中项 A : 等差中项公式 $A = \frac{9+3}{2} = 6$ 。(2) 求公差 d : 通项公式 $a_3 = a_1 + 2d$, 已知 $a_1 = 9$, $a_3 = 3: 3 = 9 + 2d \rightarrow$ 解得 $d = -3$ 。(3) 求 S_4 : 前 n 项和公式 $S_4 = \frac{4(a_1+a_4)}{2} = 2(a_1 + a_4)$; $a_4 = a_1 + 3d = 9 + 3 \times (-3) = 0$; $S_4 = 2 \times (9 + 0) = 18$ 。

英文解析 (English Explanation)

(1) Find the arithmetic mean A : Arithmetic mean formula $A = \frac{9+3}{2} = 6$ 。(2) Find the common difference d : General term formula $a_3 = a_1 + 2d$, given $a_1 = 9$ and $a_3 = 3: 3 = 9 + 2d \rightarrow$ Solve for $d = -3$ 。(3) Find S_4 : Sum of first n terms formula $S_4 = \frac{4(a_1+a_4)}{2} = 2(a_1 + a_4)$; $a_4 = a_1 + 3d = 9 + 3 \times (-3) = 0$; $S_4 = 2 \times (9 + 0) = 18$ 。